Artificial Intelligence Lab

School: Sharda School of Computing Science and Engineering
Department: Computer Science and Engineering
Course Code: CSP472
Semester: 7th
Term: 2501
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Lab Experiments

Unit	Experiments	COs	Blooms Level
Unit 1	**Practical based on goal based problems**		
	1. Introduction to Lisp, basic programming structures, and sample programs related to AI. 2. Write a LISP function to compute the sum of squares. 3. Write a LISP function to compute the difference of squares. (if x > y return x2 – y2, Otherwise y2 – x2). 4. Write a Recursive LISP function that takes one argument as a list and returns the last element of the list. (Do not use the last predicate.) 5. Write a Recursive LISP function that takes one argument as a list and returns a list except the last element of the list. (Do not use butlast. 6. Write a Recursive LISP function that takes one argument as a list and returns the reverse of the list. (Do not use reverse predicate). 7. Write a Recursive LISP function that takes two arguments: first, an atom; second, a list. It returns a list after removing the first occurrence of that atom within the list. 8. Write a Recursive LISP function that appends two lists together. 9. Write a recursive LISP function that takes 2 lists as arguments and returns a list containing alternate elements from each list. 10. Write a function that computes the factorial of a number.(Factorial of 0 is 1, and factorial of n is n*(n-1)*...1.Factorial is defined only for integers greater than or equal to 0.) 11. Write a function that evaluates a fully parenthesized infix arithmetic expression. For example, (infix (1+ (2*3))) should return 7. 12. Write a function that performs a depth-first traversal of a binary tree. The function should return a list containing the tree nodes in the order they were visited.	CO1	K1, K2, K3
Unit 2	**Practical related to uninformed search algorithm**		
	1. Write a LISP/Python program for the water jug problem. 2. Write a LISP/Python program that determines whether an integer is prime. 3. Implement BFS and DFS on a graph using Python and compare the traversal order. 4. Implement Uniform Cost Search for pathfinding in weighted graphs. Preferably in Python. 5. Simulate Depth-Limited and Iterative Deepening Search in Python. 6. Apply Greedy BFS on a map with heuristics using Python. 7. Implement A* algorithm using heuristics for shortest pathfinding. Preferably in Python. 8. Solve Sudoku using backtracking and forward-checking. Preferably in Python.	CO2	K3, K4
Unit 3	**AGENT, KNOWLEDGE & REASONING**		
	1. Simulate a logic-based agent for the Wumpus World. Preferably in Python. 2. Generate truth tables and evaluate logic expressions. Preferably in Python. 3. Create inference engines using chaining techniques. Preferably in Python.	CO3	K3

Unit 4	**APPLICATION**		
	1. Build and visualize decision trees using Python/sklearn. 2. Train a logistic regression model on a dataset. Preferably in Python. 3. Implement K-Nearest Neighbors classifier using scikit-learn. Preferably in Python. 4. Use ensemble methods for classification on real-world data. Preferably in Python. 5. Build a basic neural network with one layer in NumPy. Preferably in Python. 6. Train MLP using TensorFlow/Keras on a dataset. Preferably in Python. 7. Train CNN on MNIST/CIFAR-10 using Keras. Preferably in Python.	CO4	K3, K4, K5
Unit 5	**Practical related to knowledge representations, logical reasoning, and Course project**		
	1. Introduction to PROLOG and program structure 2. Write PROLOG program to categorize animal characteristics. And Similar type of two programs related to knowledge representation. 3. Write PROLOG program to solver for the linear equation A*X + B = 0. Let the predicate linear (A, B, X) return the root X of the equation. And Similar type of two programs. 4. Write a PROLOG program that answers questions about family members and relationships includes predicates and rules which define sister, brother, father, mother, grandchild, grandfather and uncle. The program should be able to answer queries such as the following: father(x, Amit) grandson(x, y) uncle (sumit, puneet) mother (anita, x) 5. Build a simple pattern-based chatbot using Python. 6. Perform sentiment classification using NLTK or sklearn pipelines. Preferably in Python. 7. Course Project Work Review-1: Problem Statement 8. Course Project Work Review-2: Design Specification 9.Course Project Work Review-3: Development and Results	CO5, CO6	K4, K5, K6